

Assignment 1: National Economic Vulnerability

Which national conditions are associated with youth unemployment?

Software: R

Weight: 3.33 % of the final course grade

Deadline: To be announced by the TA

Scenario

You are a junior analyst in an economic-policy unit. The unit is preparing a briefing on youth unemployment across countries and wants to know which economic, labour-market, digital and institutional conditions are associated with higher or lower rates. Your task is to prepare a reproducible empirical analysis and translate the findings into a concise briefing for a non-technical policy audience.

Decision question

Which national conditions are associated with differences in youth unemployment across countries?

Purpose

This assignment assesses your ability to understand and prepare country-level economic data, apply linear regression appropriately, interpret statistical results, and communicate evidence without making unsupported causal claims.

Assigned files

File	Purpose
country_context_raw.csv	Country-level data to be cleaned and analysed
country_context_codebook.xlsx	Definitions, units, sources, years and missing-value rules

Expected data structure

The dataset contains one row per country for a recent reference period. The assigned outcome is youth unemployment. Potential explanatory variables represent economic performance, labour-market conditions, digital access and institutional quality.

Variable group	Illustrative variables
Outcome	Youth unemployment rate

Variable group	Illustrative variables
Economic conditions	GDP per capita, GDP growth, inflation
Labour and social conditions	Female labour-force participation, urban population
Connectivity	Internet use
Institutions	Government effectiveness, political stability
Controls	Region and income group

Required tasks

Female Labour-Force Participation, Government effectiveness and Political Stability are not present in the dataset. You will have to find them and add them to the dataset yourselves, along with the Indicator, URL and all associated information just like the codebook of each tutorial.

1. Define the analytical problem

- State the decision question in your own words.
- Identify the unit of observation and data structure.
- Identify the dependent variable.
- Select a theoretically defensible set of explanatory variables.
- State the expected direction of each main relationship before estimating the model.

2. Document the data source and suitability

- Identify the original provider of every variable used.
- Report each variable's definition, unit and reference year.
- Explain whether the indicators are observed, estimated or modelled where this information is available.
- Identify at least two comparability or measurement limitations.
- Explain why the dataset is suitable for the decision question.

3. Inspect and prepare the data

- Inspect dimensions, names and variable classes.
- Identify missing values, duplicate records and implausible values.
- Explain and implement the missing-data treatment.
- Correct inappropriate variable classes.
- Create justified transformations, including the logarithm of GDP per capita where appropriate.
- Save the cleaned analytical dataset separately from the original file.

4. Conduct exploratory analysis

- Report relevant descriptive statistics.
- Examine the distribution of youth unemployment.
- Compare the outcome across regions or income groups.

- Produce at least two relationship plots.
- Identify observations that deserve further investigation without deleting them automatically.

5. Estimate the regression models

- Estimate one simple linear regression.
- Estimate at least two defensible multiple-regression specifications.
- Explain why variables are added or removed between specifications.
- Interpret coefficients using the correct units.
- Report confidence intervals and relevant hypothesis tests.
- Compare R-squared and adjusted R-squared without treating them as the only model-selection criteria.
- Examine residual plots and heteroskedasticity.
- Report robust standard errors where appropriate.

6. Interpret and communicate the evidence

- Identify the most important statistical findings.
- Discuss practical significance as well as statistical significance.
- Explain what the results may mean for economic or labour-market policy.
- Distinguish clearly between association and causality.
- Identify what additional data or research would be required before recommending an intervention.

Required submission

Item	Requirement
1. R script	Complete, organised, commented and executable from the supplied project structure
2. Clean dataset	CSV file created through the submitted script; do not overwrite the raw file
3. A report including:	Maximum four pages, excluding references and appendix
Policy summary	Maximum one page for a non-technical decision-maker
Regression table(s)	Clearly labelled tables and comparing the main specifications
Figures	Final figures with titles, labels, units and source notes
4. AI log	

Recommended report structure

1. Decision question and economic reasoning
2. Data, variables and source suitability
3. Exploratory evidence
4. Regression approach and results

5. Policy interpretation
6. Limitations and next evidence needed

Policy summary requirements

The one-page policy summary must answer:

- What was investigated?
- What are the two or three most important findings?
- Why might they matter for a policymaker?
- How confident should the reader be?
- What should not be concluded?
- What evidence should be collected next?

Minimum technical requirements

- All analysis must be reproducible by running the submitted R **script**.
- Use **relative file paths**. Code tied to a personal computer path will not be considered fully reproducible.
- Do not manually edit the cleaned CSV after it is produced by the script.
- All tables and figures must be generated from the submitted data and code.
- Every figure must include a title, labelled axes, units and a source note.
- Keep the original variable meanings and units visible in the codebook or report.

Interpretation standards

Your language must distinguish among the following:

Evidence level	Acceptable wording
Descriptive	Youth unemployment is higher in the observed group.
Associational	The variable is associated with youth unemployment after controlling for the included factors.
Predictive	The model predicts higher youth unemployment for observations with these characteristics.
Causal	Changing the factor would change youth unemployment. This claim requires a valid causal design.

Assessment rubric

Criterion	Marks	What is assessed
Question, reasoning and data understanding	4	Clear question, appropriate variables, source documentation and understanding of the unit of analysis
Data preparation and exploratory analysis	4	Reproducible cleaning, appropriate summaries, visualisations and justified decisions
Regression analysis	6	Correct estimation, specification, diagnostics, inference and coefficient interpretation
Policy communication and limitations	4	Decision relevance, clarity, practical significance and disciplined claims
Reproducibility and presentation	2	Executable code, organised files, complete labels, citations and professional presentation
Total	20	Converted to 3.33% of the final course grade

Submission folder

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BUAD405_Applied_Econometrics/
├── data/
├── scripts/
├── output/
├── submissions/
│   └── assignments/
│       ├── assignment01/
│       │   ├── assignment01_name_ID.R
│       │   ├── clean_dataset.csv
│       │   ├── assignment01_report.pdf
│       │   └── assignment01_AI_log.pdf
│       └── assignment02/
│           ...

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Academic integrity and responsible analysis

- Follow the university and course rules for collaboration, code use and generative AI.
- Acknowledge any external code, tutorial or analytical assistance used.
- You remain responsible for every analytical choice, line of code and interpretation submitted.
- Do not alter, hide or selectively remove observations merely to obtain preferred results.

Final submission checklist

- The raw dataset remains unchanged.
- The script runs from beginning to end.
- The clean dataset is produced by the script.
- All variables are defined and cited.
- Every coefficient is interpreted in the correct unit.
- The report separates association from causation.
- Figures and tables are numbered, labelled and sourced.
- The policy summary is understandable without reading the R output.
- All required files to be submitted in a folder named submissions in the course project file.